

ScribSD+: Scribble-supervised medical image segmentation based on simultaneous multi-scale knowledge distillation and class-wise contrastive regularization

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ABSTRACT

Despite that deep learning has achieved state-of-the-art performance for automatic medical image segmentation, it often requires a large amount of pixel-level manual annotations for training. Obtaining these high-quality annotations is time-consuming and requires specialized knowledge, which hinders the widespread application that relies on such annotations to train a model with good segmentation performance. Using scribble annotations can substantially reduce the annotation cost, but often leads to poor segmentation performance due to insufficient supervision. In this work, we propose a novel framework named as ScribSD+ that is based on multi-scale knowledge distillation and class-wise contrastive regularization for learning from scribble annotations. For a student network supervised by scribbles and the teacher based on Exponential Moving Average (EMA), we first introduce multi-scale prediction-level Knowledge Distillation (KD) that leverages soft predictions of the teacher network to supervise the student at multiple scales, and then propose class-wise contrastive regularization which encourages feature similarity within the same class and dissimilarity across different classes, thereby effectively improving the segmentation performance of the student network. Experimental results on the ACDC dataset for heart structure segmentation and a fetal MRI dataset for placenta and fetal brain segmentation demonstrate that our method significantly improves the student's performance and outperforms five state-of-the-art scribble-supervised learning methods. Consequently, the method has a potential for reducing the annotation cost in developing deep learning models for clinical diagnosis.

1. Introduction

With the advancement of medical imaging technology and its wider application, automatic medical image segmentation plays an important role for accurate computer-assisted diagnosis of disease and treatment planning. Convolutional Neural Networks (CNNs) (Milletari et al., 2016; Çiçek et al., 2016; Chen et al., 2021a) have achieved great success in automatic image segmentation. Additionally, Transformer-based networks such as TransUNet (Chen et al., 2021b; Zou et al., 2023) have also demonstrated high performance by leveraging self-attention mechanisms. However, most of these methods require large-scale images with accurate pixel-level dense annotations for model training (Isensee et al., 2021; Kushnure and Talbar, 2021; Ammar et al., 2021). Unfortunately, medical image annotation is a time-consuming, expensive, and expert-dependent process, making obtaining large-scale annotated datasets difficult.

In order to reduce the annotation cost of model training, researchers are actively exploring more annotation-efficient methods. Weakly Supervised Learning (WSL) uses sparse annotations to train the model (Valvano et al., 2021; Dolz et al., 2021; Dorent et al., 2021), avoiding the need for dense annotations. Weak annotations such as image-level labels (Wu et al., 2019), bounding boxes (Rajchl et al., 2016; Ou et al., 2023), point annotations (Zhai et al., 2023), and scribbles (Lin et al., 2016; Valvano et al., 2021) are more accessible to obtain than pixel-level dense annotations, and can effectively reduce the annotation cost. Compared with image-level labels, bounding boxes and point annotations, scribbles are more flexible to annotate targets with various shapes and often lead to better performance (Lin et al., 2016). Therefore, this work focuses on exploring using scribble annotations to efficiently and robustly train high-performance medical image segmentation models.

An intuitive challenge is that although scribbles can flexibly annotate targets of various shapes, the sparse annotations still lack sufficient

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structural information, as illustrated in Fig. 1. Most existing methods require pseudo labels to generate supervision signals for unlabeled pixels, which easily leads to (i) poor performance of the learned model due to that the pseudo labels may be inaccurate in the presence of large amounts of noise; (ii) high uncertainty in boundary regions since the scribbles are often given in the central region of a target, leading to ambiguous distribution of different classes at the boundary. Directly using pseudo labels without considering the noise may limit the model's performance, and failing to deal with the boundary regions could lead to inaccurate segmentation.

To deal with these problems, we propose an efficient scribble-based learning method called ScribSD+ that is based on multi-scale Knowledge Distillation (KD) and class-wise contrastive regularization for medical image segmentation based on scribble annotations. The contributions are three-fold. Firstly, we introduce a KD-based method for robustly learning from scribbles for segmentation. Compared with existing works using hard pseudo labels (Luo et al., 2022a), we introduce Multi-scale Prediction Distillation (Ms-PD) that uses soft predictions obtained by a teacher to supervise the student at multiple scales, as soft predictions are more noise-robust, while hard labels tend to contain more noise (Xu et al., 2020). Secondly, in order to reduce ambiguous predictions at the boundary and enhance class-wise feature distinctiveness, we introduce Multi-scale class-wise Contrastive Regularization (Ms-CR) that uses contrastive loss to encourage the feature vectors of the same class from the teacher and student to be close and those of different classes to be far apart. Multi-scale design helps to better transfer knowledge at different resolution levels from the teacher to student. In addition, compared with classic KD methods (Hinton et al., 2015; Xu et al., 2020) that require pre-training a teacher model, we use self-distillation that efficiently generates the teacher on-the-fly based on the Exponential Moving Average (EMA) of model parameters.

This paper is a substantial extension of our preliminary work ScribSD published in MICCAI 2023 MILLanD Workshop (Qu et al., 2023), where we proposed prediction distillation based on soft predictions to deal with noisy pseudo labels and multi-scale feature distillation that transfers the high-quality knowledge contained in the teacher model to the student, and validated its effectiveness on a fetal brain and placenta segmentation dataset. In this extension, we give a more detailed description and analysis of the multi-scale prediction distillation, and replace our previous multi-scale feature distillation (Qu et al., 2023) by multi-scale class-wise contrastive regularization, where the latter can learn more distinctive feature representations for different classes for better performance while supervising the student by the teacher at the feature level. Additionally, our method is further validated on the public ACDC dataset for heart structure segmentation. Compared with learning only from the scribbles with partial Cross-Entropy on these two datasets, our method improved the average Dice by around 6.0 and 15.5 percentage points, respectively, and it significantly outperformed five state-of-the-art weakly supervised methods.

2. Related works

2.1. Weakly-supervised learning

To learn from scribble annotations for segmentation, existing methods are mainly based on introducing regularization or pseudo labels for unlabeled pixels. For regularization-based methods, Lin et al. (2016) built a superpixel-based graphical model trained by propagating information from scribbles to unlabeled pixels. Kim and Ye (2019) proposed a level-set-based Mumford–Shah function to regularize the segmentation results. Kervadec et al. (2019) constrained the volume of the predicted mask by introducing higher-order inequalities in the loss function and employed deep tightness prior for learning from bounding box annotations. Valvano et al. (2021) proposed a learning method based on multi-scale adversarial attention gates for raining a segmentation

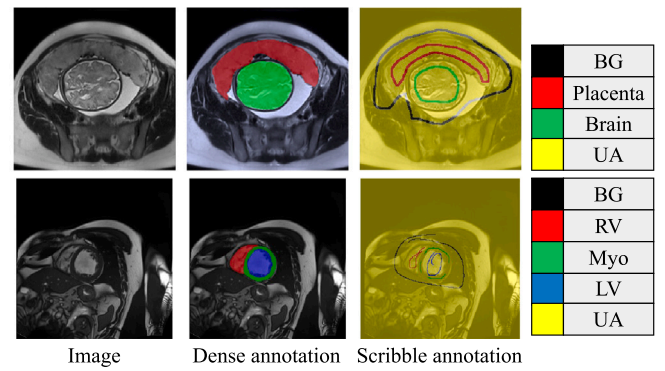


Fig. 1. Examples of dense and scribble annotations. RV, Myo and LV represent the right ventricle, myocardium, left ventricle respectively. BG and UA represent the background and unannotated pixels respectively.

model with scribble annotations. Scribble2D5 (Chen and Hong, 2022) expands the scribbles based on super-pixels and uses active contours to constrain the prediction results. ShapePU (Zhang and Zhuang, 2022) combines positive-unlabeled learning and global consistency regularization to obtain supervisory information from unlabeled pixels to train a heart segmentation model. Conditional Random Field (CRF) (Tang et al., 2018b) loss and normalized cut loss (Tang et al., 2018a) have also been proposed for regularization. Additionally, pseudo-label-based methods utilize weak annotations to generate pseudo labels for network training. Scribble2Label (Lee and Jeong, 2020) generates pseudo labels based on the EMA of the predictions and filters out low-quality labels for robust learning. Luo et al. (2022a) proposed Dynamically Mixed Pseudo Label Supervision (DMPLS) where the predictions of two decoders are mixed randomly to serve as pseudo labels. The Uncertainty-aware Self-ensembling and Transformation-consistent Mean Teacher Model (USTM) (Liu et al., 2022) leverages an uncertainty-aware mean teacher to generate pseudo labels for supervising the student based on a Mean Square Error (MSE) loss. These attempts have been effective to a certain extent, however, regularization-based methods often face performance limitations due to their inability to fully capture complex patterns in the data, potentially leading to suboptimal model performance. For pseudo-label-based methods, since the generated pseudo labels are inaccurate under the influence of noise, directly using them as supervision signals may limit the performance of the model.

2.2. Contrastive learning

As a self-supervised learning method, contrastive learning usually adopts a contrastive loss to force the representations of positive pairs to be close and those of negative pairs to be far away to learn feature representations (Hadsell et al., 2006). Previous contrastive learning methods were mainly used for self-supervised pre-training, with the aim of training a powerful and representative feature extractor that can distinguish between similar and dissimilar samples, so that it can be fine-tuned for downstream tasks (He et al., 2020; Chen et al., 2020). Recently in the fields of computer vision and medical image analysis, contrastive learning has also been applied to annotation-efficient learning (Zhao et al., 2024). For example, Kang et al. (2019) proposed a contrastive adaptation network to achieve class-aware domain alignment by minimizing the intra-class distance and maximizing the inter-class distance. Chaitanya et al. (2020) proposed a combination of global contrastive loss and local contrastive loss for semi-supervised 3D medical image segmentation. Lei et al. (2021) proposed contrastive learning of relative position regression for one-shot target localization of 3D medical images. You et al. (2022) proposed a momentum-contrastive voxel-level representation learning method for semi-supervised medical image segmentation by combining low-level

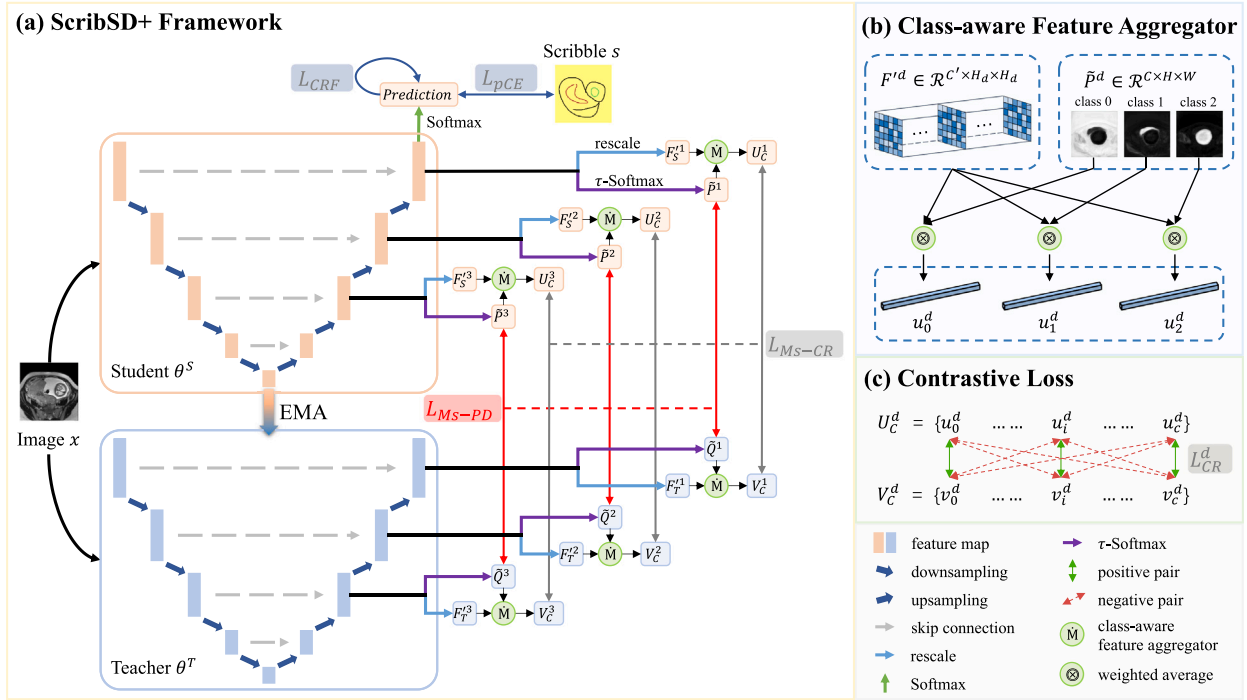


Fig. 2. The proposed ScribSD+ framework based on multi-scale knowledge distillation and class-wise contrastive regularization. Multi-scale Prediction Distillation (L_{MS-PD}) based on soft labels and Multi-scale class-wise Contrastive Regularization (L_{MS-CR}) are combined for supervising the student on unannotated pixels.

and high-level contrastive objectives with consistency loss. Gu et al. (2022) proposed a contrastive semi-supervised learning method for medical image segmentation, using a cross-domain contrastive learning strategy to extract domain-invariant features. Zhou et al. (2023) learned prototypes of each class and encouraged the similarity between the class-wise feature to its prototypes. However, it only encourages the compactness of each class, while the dissimilarity of difference classes is not considered. Differently from this work, we design a novel class-wise contrastive regularization strategy without complex prototype learning process for scribble-based learning in segmentation tasks. Our method simultaneously encourages intra-class compactness and inter-class distinctness between the features of teacher and student models.

3. Method

Our proposed ScribSD+ based on multi-scale knowledge distillation and class-wise contrastive regularization for learning from scribble annotations is illustrated in Fig. 2. For a student model supervised by the scribbles and a regularization loss based on CRF, we generate a teacher model based on EMA for self-distillation, where multi-scale prediction-level distillation and class-wise contrastive regularization are combined for the student to better learn segmentation information and feature representations from the teacher.

3.1. Student model with scribble supervision

In this work, we consider a UNet-like (Ronneberger et al., 2015) structure for segmentation due to its high performance in a wide range of tasks. For a training set with scribble annotations, a partial Cross-Entropy (pCE) loss is used to learn from the scribbles:

$$L_{pCE} = -\frac{1}{|\Omega_S|} \sum_{i \in \Omega_S} \sum_{c=0}^C S_i^c \log(P_i^c) \quad (1)$$

where C denotes the class number for segmentation. S is the one-hot scribble annotation map, Ω_S is the set of labeled pixels, and P is

the Softmax-based probability map predicted by the student model. S_i^c and P_i^c are probabilities of the i th pixel being class c in the scribble annotation and network prediction, respectively.

Since just applying L_{pCE} on the sparse annotations usually leads to inadequate supervision, especially around boundaries (Obukhov et al., 2019), we further use a CRF loss (L_{CRF}) to regularize the network output, which encourages prediction consistency between neighboring pixels with similar features:

$$L_{CRF} = \sum_{c=0}^C \sum_{(i,j) \in \Phi} G(i,j) P_i^c (1 - P_j^c) \quad (2)$$

where Φ denotes the set of pixel pairs in the image. $G(i,j)$ refers to the Gaussian kernel bandwidth filter:

$$G(i,j) = \frac{1}{\omega} \exp \left(-\frac{\|H_i - H_j\|^2}{2\sigma_H^2} - \frac{\|I_i - I_j\|^2}{2\sigma_I^2} \right) \quad (3)$$

where H_i and I_i denote the spatial position and the intensity value of pixel i , respectively. ω is the weight for normalization, σ_H and σ_I are hyper-parameters to control the scale of Gaussian kernels.

3.2. Scribble-supervised self-distillation

Instead of pre-training an extra teacher model offline for KD (Hinton et al., 2015; Xu et al., 2020), we obtain the teacher model during the training process on the fly, where the teacher has the same structure as the student model, but with different model weights. As shown in Fig. 2(a), let θ^S and θ^T denote the weights of the student and teacher models, respectively. The teacher is updated as an EMA of the student: $\theta_t^T = \lambda \theta_{t-1}^T + (1 - \lambda) \theta_t^S$, where θ_t^S and θ_{t-1}^T are weights of the teacher and student at iteration t and $t-1$, respectively. $\lambda = \min(1 - \frac{1}{t+1}, 0.99)$ is the EMA momentum that gradually increases to 0.99, where t denotes the current number of iterations.

3.2.1. Multi-scale prediction distillation

When sparse scribble annotations are used to generate pseudo labels, the quality of the pseudo labels is usually poor due to the presence

of a large amount of noise, and using low-quality pseudo labels may limit training of the student model. To address this problem, we propose Multi-scale Prediction Distillation (Ms-PD). We use high-quality soft predictions obtained by the teacher at multiple scales to supervise the student, as soft predictions are more robust to noise, while hard labels tend to contain more noise (Xu et al., 2020). In addition, by using supervised training at multiple scales, the pseudo labels generated by the high-performance teacher model can be better utilized, and the student model can learn multi-level and rich information.

Specifically, we obtain soft predictions for the teacher and student at D scales in the decoder. For scale d , let $F^d \in \mathcal{R}^{C_d \times H_d \times W_d}$ denote the last feature map in the decoder, where C_d is the channel number and $H_d \times W_d$ is the spatial dimension. We use 1×1 convolution to convert the channel number to C and then obtain soft predictions based on a temperature-calibrated Softmax (τ -Softmax) that are more noise-tolerant (Müller et al., 2019). Let $\tilde{P}^d$ and $\tilde{Q}^d$ denote the soft predictions at the d th scale for the student and teacher, respectively. Let $\tilde{Q}_{c,n}^d$ denote the probability value in the c th channel at the n th pixel of $\tilde{Q}^d$. It is defined as:

$$\tilde{Q}_{c,n}^d = \frac{\exp(z_{c,n}^d / \tau)}{\sum_c \exp(z_{c,n}^d / \tau)} \quad (4)$$

where $z_{c,n}^d$ is the logit output on the c th channel at the n th pixel obtained at scale d . $\tau \geq 1$ is a hyper-parameter to control the softness, where a larger τ value leads to a softer output, i.e., the probability for each class is less likely to be the extreme values of 0.0 or 1.0. Note that $\tau = 1$ corresponds to the standard Softmax operation. Considering that as the training progresses, the performance of the teacher model gradually improves and the impact of noise on the quality of pseudo labels gradually becomes smaller, we decrease the value of τ from a maximum value τ_{max} to a minimum value τ_{min} according to the cosine ramp down function, which is expressed as:

$$\tau = \max(\tau_{min}, \tau_{max} \cdot \frac{1}{2} (\cos(\frac{\pi t}{t_{max}}) + 1)) \quad (5)$$

where t denotes the current iteration number and t_{max} is the maximum iteration number for ramp-down.

Based on the teacher's soft prediction $\tilde{Q}^d$ and student's soft prediction $\tilde{P}^d$ at scale d , we use KL-Divergence to encourage the class distribution obtained by the student to be close to that of the teacher:

$$L_{PD}^d = \frac{1}{N} \sum_{n=0}^N \sum_{c=0}^C \tilde{Q}_{c,n}^d \log(\tilde{Q}_{c,n}^d / \tilde{P}_{c,n}^d) \quad (6)$$

where N denotes the number of pixels in the image. The multi-scale prediction-level distillation loss L_{Ms-PD} is defined as:

$$L_{Ms-PD} = \sum_{d=1}^D \gamma_d \cdot L_{PD}^d \quad (7)$$

where γ_d is the weight at scale d , and the sum of γ_d at different scales is 1.0. The value of d ranges from 1 to D . Note that $d = 1$ means we only use the single-scale soft predictions at the end of decoder. A larger d value means a lower resolution that is closer to the bottleneck.

3.2.2. Multi-scale class-wise contrastive regularization

In addition to Ms-PD which focuses on prediction-level distillation at multiple scales, we also consider feature-level distillation for improving the student's performance. In our preliminary work ScribSD (Qu et al., 2023), we directly maximized the similarity between the student's features and the teacher's features. However, that method does not regularize the features' compactness and distinctiveness. To deal with this problem, we propose Multi-scale class-wise Contrastive Regularization (Ms-CR) for better feature representations. As shown in Fig. 2(b), we first use a class-aware feature aggregator to obtain class-wise features based on feature maps and the corresponding soft predictions, then use a contrastive loss to enhance the similarity of the same class and dissimilarity of different classes between the teacher

and student, so that the model can learn more distinctive multi-class feature distributions under sparse scribble annotations.

Following Section 3.2.1, we apply an extra 1×1 convolution to $F^d \in \mathcal{R}^{C_d \times H_d \times W_d}$ to convert its channel number to a fixed value of C' , and the result is denoted as $F'^d \in \mathcal{R}^{C' \times H_d \times W_d}$. For the student network, we obtain the class-wise feature at scale d from the feature map corresponding to each class by weighted average:

$$u_c^d = \frac{\sum_{n=0}^N F_n'^d \tilde{P}_{c,n}^d}{\sum_{n=0}^N \tilde{P}_{c,n}^d} \quad (8)$$

where $u_c^d \in \mathcal{R}^{C'}$ is the average feature vector for class c at scale d , and $F_n'^d \in \mathcal{R}^{C'}$ is the feature vector at pixel n in F'^d . Similarly, for the teacher network, we use $\tilde{Q}^d$ to obtain the class-wise feature and denote it as v_c^d . Let i and κ to denote two class indexes. As shown in Fig. 2(c), for the same class (i.e. $i = \kappa$), we treat (u_i^d, v_κ^d) as a positive pair. For different classes (i.e. $i \neq \kappa$), (u_i^d, v_κ^d) is considered as a negative pair. For contrastive regularization at scale d , we encourage class-wise features in a positive pair to be close and those in a negative pair to be far apart to learn more compact and distinctive class-wise feature representations.

$$L_{CR}^d = \frac{1}{C} \sum_{i=0}^C -\log \frac{\exp(\text{sim}(u_i^d, v_i^d) / \rho)}{\sum_{\kappa=0}^C \exp(\text{sim}(u_i^d, v_\kappa^d) / \rho)} \quad (9)$$

where $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity between two feature vectors, ρ is a hyper-parameter and it is set to 0.07 following He et al. (2020). The multi-scale class-wise contrastive regularization loss L_{Ms-CR} is defined as:

$$L_{Ms-CR} = \sum_{d=1}^D \gamma_d \cdot L_{CR}^d \quad (10)$$

Following Eq. (7), γ_d is the weight at scale d and the sum of γ_d at different scales is 1.0.

3.3. The overall loss function

Based on L_{Ms-PD} and L_{Ms-CR} , the overall loss for student training is:

$$L = L_{pCE} + \beta_i (\alpha \cdot L_{CRF} + L_{Ms-PD} + L_{Ms-CR}) \quad (11)$$

where α is the weight for L_{CRF} , following Obukhov et al. (2019), and based on the result of the validation set, α is set to 0.01 in our method. β_i is a weight for the ramp-up terms. Since the teacher network uses the EMA weights of successive student networks, the performance of the teacher model is poor at the early stage of training. As the training progresses, the teacher's performance gradually improves, which is leveraged by a larger β_i . Therefore β_i ramps up from 0 to its maximum value β according to sigmoid ramp up function $\beta_i = \beta * e^{-5(1-t/t_{max})^2}$, where t denotes the current iteration number and t_{max} is the maximum iteration number for ramp-up.

4. Experiment and results

4.1. Dataset

Fetal MRI dataset. MRI scans of 88 fetuses in the second trimester were collected by Single Shot Fast Spin Echo (SSFSE). The image size is 256×256 , and each volume has 25–100 slices with pixel size 0.60–1.24 mm and slice thickness 3.00–7.15 mm. Each scan has annotations for two structures, i.e., the placenta and brain. The images used in this work were manually annotated by a Radiologist. We used scribbles for the training set and pixel-level annotations for the validation and testing sets. In scribble annotations, the number of labeled foreground pixels approximately accounts for 6% of the total foreground pixels in the original dense annotations. The images were randomly split into 40, 10, and 38 scans for training, validation, and testing, respectively.

Table 1

Comparison between our method and existing weakly-supervised methods with the same backbone of UNet (Ronneberger et al., 2015) on the Fetal MRI dataset.

Method	Dice (%) $\uparrow$			ASSD (mm) $\downarrow$		
	Placenta	Brain	Average	Placenta	Brain	Average
pCE	77.73 $\pm$ 9.25	87.06 $\pm$ 5.48	82.40 $\pm$ 6.14	5.72 $\pm$ 4.26	3.63 $\pm$ 2.18	4.67 $\pm$ 2.45
MLoss (Kim and Ye, 2019)	76.87 $\pm$ 8.78	86.76 $\pm$ 6.09	81.81 $\pm$ 6.17	5.89 $\pm$ 4.37	3.52 $\pm$ 1.91	4.71 $\pm$ 2.52
TV (Luo et al., 2022b)	76.61 $\pm$ 9.51	87.25 $\pm$ 5.96	81.93 $\pm$ 6.45	5.98 $\pm$ 4.33	3.30 $\pm$ 1.93	4.64 $\pm$ 2.44
USTM (Liu et al., 2022)	80.92 $\pm$ 6.62	88.78 $\pm$ 4.84	84.85 $\pm$ 4.55	4.42 $\pm$ 3.22	3.07 $\pm$ 2.04	3.74 $\pm$ 2.00
EM (Grandvalet and Bengio, 2004)	76.92 $\pm$ 19.47	89.34 $\pm$ 4.39	83.13 $\pm$ 10.34	10.37 $\pm$ 22.33	2.58 $\pm$ 1.29	6.47 $\pm$ 11.16
DMPLS (Luo et al., 2022a)	78.18 $\pm$ 7.15	86.94 $\pm$ 7.10	82.56 $\pm$ 5.53	7.25 $\pm$ 5.80	2.84 $\pm$ 1.30	5.04 $\pm$ 3.03
ScribSD (Qu et al., 2023)	82.91 $\pm$ 8.87	92.46 $\pm$ 3.94	87.68 $\pm$ 5.49	3.68 $\pm$ 2.96	1.48 $\pm$ 0.88	2.58 $\pm$ 1.70
ScribSD+	83.81 $\pm$ 9.31*	92.95 $\pm$ 3.29*	88.38 $\pm$ 5.84*	3.37 $\pm$ 3.19	1.36 $\pm$ 0.84*	2.37 $\pm$ 1.74*

* Denotes significantly higher performance than the existing methods (p -value < 0.05 based a paired t -test).

ACDC dataset (Bernard et al., 2018). 200 short-axis cine-MRI scans from 100 patients were acquired in breath hold with a retrospective or prospective gating and with a Steady-state free precession (SSFP) sequence in short axis orientation. Each patient has two scans that are for end-diastolic (ED) and end-systolic (ES) phases, respectively. And each scan has annotations for three structures, i.e., the right ventricle (RV), myocardium (Myo), and left ventricle (LV). Each volume has 6–18 slices with pixel size 0.70–1.20 mm and slice thickness 5.00–10 mm. The minimum slice size is 154×154 and the maximum is 428×512 . The scribbles used in this work were manually annotated by Valvano et al. (2021), and the number of labeled foreground pixels accounts for around 12% of all foreground pixels in the original dense annotations. The images were split at patient level into 70%, 10% and 20% for training, validation and testing. Scribbles were used for the training set and pixel-level annotations were used for the validation and testing sets.

4.2. Implementation details

The experiments were conducted on a Ubuntu desktop with PyMIC (Wang et al., 2023) and an NVIDIA GeForce RTX 3090Ti GPU. For these two datasets, due to the large slice thickness, we employed 2D CNNs for slice-level segmentation and stacked the results in a volume for 3D evaluation. 2D UNet (Ronneberger et al., 2015) was used for segmentation, with Adam optimizer for training. Dropout was used in the two low-resolution levels and the dropout ratio was 0.5. For both datasets, Z-score Normalization was applied during preprocessing. For the Fetal MRI dataset, random flipping and random cropping were used for data augmentation. The learning rate was initialized as 0.001 and reduced by half for every 150 epochs. The batch size was 4 and the total epoch number was 800. The maximum iteration number for ramp-up t_{max} was 5000. For the ACDC dataset, we used random flipping and random cropping for data augmentation. The ReduceLROnPlateau learning rate strategy was used to adjust the learning rate online. The batch size was 4 and the total epoch number was 600. The maximum iteration number for ramp-up t_{max} was 15000. Based on the best performance on the validation set, the hyper-parameter setting was: $\tau_{max} = 5$, $\tau_{min} = 2$, $\alpha = 0.01$, $C' = 16$, $\beta = 1$, $D = 3$, $\gamma_1 = 0.5$, $\gamma_2 = 0.3$, and $\gamma_3 = 0.2$. Following Obukhov et al. (2019), the parameters for CRF loss were $\omega = 1$, $\sigma_H = 5$ and $\sigma_I = 0.1$. For quantitative evaluation of the 3D segmentation results, we measured the Dice score and Average Symmetric Surface Distance (ASSD).

4.3. Comparison with state-of-the-art methods

Our method was compared with seven existing methods: (1) only pCE, (2) Dynamically Mixed Pseudo Labels Supervision (DMPLS) (Luo et al., 2022a), (3) Entropy Minimization (EM) (Grandvalet and Bengio, 2004), (4) Mumford–Shah Loss (MLoss) (Kim and Ye, 2019), (5) Total Variation (TV) (Luo et al., 2022b) for regularization, (6) USTM (Liu et al., 2022) and (7) ScribSD that denotes our previous method (Qu et al., 2023). Note that ScribSD uses prediction distillation only for the highest resolution level, i.e., single-scale, and it maximizes the feature similarity between the teacher and the student, rather than using class-wise contrastive regularization.

4.3.1. Results on the fetal MRI dataset

We compared our method with the above-mentioned methods on the fetal MRI dataset, and the results are shown in Table 1. Using pCE loss only obtained an average Dice of 77.73% and 87.06% for the placenta and fetal brain, respectively. Our method achieved the best performance and improved it to 83.81% and 92.95% for the two classes, respectively. Compared with USTM (Liu et al., 2022), our method significantly improved the average Dice by 3.53 percent points and reduced the ASSD by 1.37 mm, with p -value < 0.05 according to a paired t -test. Fig. 3 presents a visual comparison of segmentation results, where red and green colors show the placenta and fetal brain, respectively. It shows that the segmentation result obtained by our proposed method was the closest to the ground truth. In contrast, the other methods obtained more under-segmented or over-segmented regions, as highlighted by yellow arrows. This is due to the design of our Ms-PD and Ms-CR, which enables the network to learn fine class-wise feature information, thereby improving the distinctiveness of different classes for better performance.

4.3.2. Results on the ACDC dataset

Table 2 presents a quantitative comparison on the ACDC dataset. Using pCE loss, the average Dice score was only 72.76% and the ASSD is as high as 38.78 mm, mainly due to a lot of over-segmentations. Our method achieved the best performance on all two evaluation metrics, improving the average Dice score to 88.45% and decreasing the average ASSD to 2.52 mm, significantly outperforming other methods. Compared with DMPLS (Luo et al., 2022a), our method significantly improved the average Dice by 2.03 percent points and reduced the ASSD by 3.76 mm. It can be seen from the visual comparison of segmentation results shown in Fig. 4 that the segmentation result obtained by our proposed method was more accurate in the boundary regions and was the closest to the ground truth. The segmentation results of other methods suffered from over-segmentation or under-segmentation, especially in boundary regions.

4.4. Ablation study

We introduced the multi-scale prediction distillation and class-wise contrastive regularization strategies in the network to improve segmentation performance in the proposed method. This section will first introduce ablation experiments to evaluate the effectiveness of different supervision strategies on our framework. Then, we explored the sensitivity of hyper-parameters D , τ_{max} , τ_{min} , β respectively. All experiments were performed on the fetal MRI dataset.

4.4.1. Effectiveness of different supervision strategies

To investigate the effectiveness of each of the supervision terms used in our method, we set pCE as the baseline, and gradually added the CRF loss, Ms-PD loss, and Ms-CR loss, respectively. The PD loss and CR loss refer to single-scale (for the highest resolution level only) prediction distillation and contrastive regularization respectively. The PD loss was also compared with an MSE loss applied to the probability

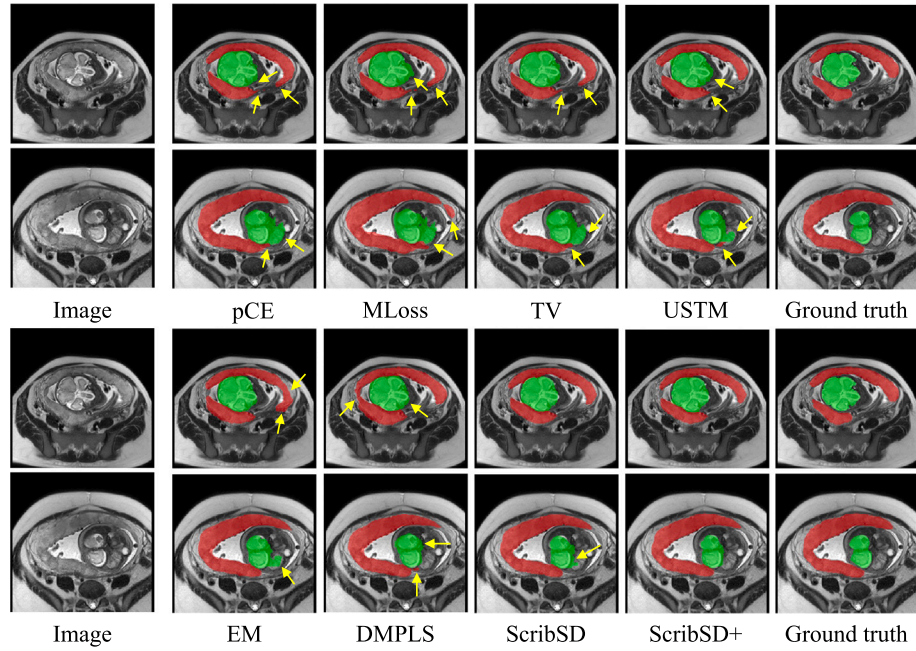


Fig. 3. Qualitative comparison of our proposed method and several existing methods on the fetal MRI dataset.

Table 2

Comparison between our method and existing weakly-supervised methods with the same backbone of UNet (Ronneberger et al., 2015) on the ACDC dataset.

Method	Dice (%)↑				ASSD (mm)↓			
	RV	Myo	LV	Average	RV	Myo	LV	Average
pCE	79.20 ± 10.16	64.95 ± 10.27	74.12 ± 16.60	72.76 ± 8.73	20.02 ± 23.88	46.47 ± 18.18	49.84 ± 27.21	38.78 ± 15.03
MLoss (Kim and Ye, 2019)	69.87 ± 9.73	62.22 ± 8.58	72.09 ± 15.83	68.06 ± 8.86	43.96 ± 22.77	45.41 ± 18.39	53.56 ± 27.01	47.64 ± 18.10
TV (Luo et al., 2022b)	79.00 ± 9.40	75.22 ± 7.28	87.91 ± 6.78	80.71 ± 5.97	23.06 ± 16.57	22.79 ± 13.07	15.31 ± 14.38	20.39 ± 11.89
USTM (Liu et al., 2022)	84.99 ± 7.37	76.77 ± 6.83	88.22 ± 6.12	83.33 ± 4.15	7.48 ± 5.04	19.82 ± 13.28	10.15 ± 10.90	12.48 ± 7.94
EM (Grandvalet and Bengio, 2004)	85.22 ± 7.30	81.76 ± 5.83	89.75 ± 6.50	85.58 ± 4.98	7.93 ± 9.98	8.81 ± 8.50	8.82 ± 12.48	8.52 ± 8.88
DMPLS (Luo et al., 2022a)	86.44 ± 6.91	82.22 ± 6.23	90.62 ± 6.32	86.42 ± 4.39	4.74 ± 4.39	7.51 ± 9.08	6.61 ± 8.54	6.28 ± 6.63
ScribSD (Qu et al., 2023)	86.62 ± 6.78	84.02 ± 5.96	92.05 ± 4.20	87.56 ± 3.77	3.88 ± 5.83	5.05 ± 9.03	4.01 ± 7.64	4.31 ± 6.55
ScribSD+	87.33 ± 6.54	85.39 ± 5.53*	92.64 ± 4.23*	88.45 ± 3.97*	2.56 ± 3.03*	2.63 ± 7.68*	2.37 ± 6.61*	2.52 ± 5.47*

* Denotes significantly higher performance than the existing methods (p -value < 0.05 based a paired t-test).

Table 3

Ablation study of different supervision strategies on the fetal MRI dataset. PD and CR refer to single-scale prediction distillation and contrastive regularization respectively. MSE corresponds to the typical mean teacher method.

Loss functions							Dice (%)↑			ASSD (mm)↓		
pCE	CRF	MSE	PD	CR	Ms-PD	Ms-CR	Placenta	Brain	Average	Placenta	Brain	Average
✓							81.24 ± 6.45	83.81 ± 9.46	82.53 ± 6.37	4.83 ± 2.71	7.07 ± 8.92	5.95 ± 4.64
✓	✓						79.34 ± 11.10	88.26 ± 5.28	83.80 ± 6.14	9.07 ± 12.24	2.62 ± 1.33	5.85 ± 6.00
✓	✓	✓					79.87 ± 11.36	87.15 ± 7.29	83.51 ± 6.49	8.04 ± 12.55	2.77 ± 1.95	5.40 ± 6.09
✓	✓		✓				83.66 ± 4.76	87.53 ± 6.03	85.59 ± 4.31	3.75 ± 1.56	2.70 ± 1.42	3.22 ± 0.96
✓	✓			✓			86.38 ± 3.96	90.15 ± 6.13	88.27 ± 4.34	3.18 ± 1.46	2.45 ± 1.90	2.81 ± 1.23
✓	✓				✓	✓	86.60 ± 3.59	92.16 ± 3.59	89.38 ± 3.02	3.03 ± 1.40	1.64 ± 0.86	2.33 ± 0.86

Table 4

Ablation study of different setting of hyper-parameter D and the corresponding γ_d on the fetal MRI dataset.

D	γ_d					Dice (%)↑			ASSD (mm)↓		
	γ_1	γ_2	γ_3	γ_4	γ_5	Placenta	Brain	Average	Placenta	Brain	Average
1	1					86.38 ± 3.96	90.15 ± 6.13	88.27 ± 4.34	3.18 ± 1.46	2.45 ± 1.90	2.81 ± 1.23
2	0.7	0.3				85.63 ± 4.44	90.96 ± 5.56	88.29 ± 4.07	3.16 ± 1.51	1.92 ± 1.35	2.54 ± 0.90
3	0.5	0.3	0.2			86.60 ± 3.59	92.16 ± 3.59	89.38 ± 3.02	3.03 ± 1.40	1.64 ± 0.86	2.33 ± 0.86
4	0.4	0.3	0.2	0.1		85.96 ± 4.02	90.79 ± 5.63	88.38 ± 4.08	2.97 ± 1.28	1.95 ± 1.36	2.46 ± 0.86
5	0.3	0.3	0.2	0.1	0.1	85.14 ± 5.05	90.77 ± 5.33	87.96 ± 4.55	3.25 ± 1.51	1.91 ± 1.20	2.58 ± 0.96

predictions obtained by standard Softmax, which corresponds to the typical mean teacher method (Tarvainen and Valpola, 2017). The results of different settings of the training loss on the validation set are shown in Table 3. It can be observed that pCE only obtained an average

Dice of 82.53% and introducing the CRF loss improved it to 83.80%. Applying single-scale prediction distillation (PD) further improved it to 85.59%, while replacing PD with MSE led to a lower performance (83.51%). On this basis, applying single-scale contrastive regularization

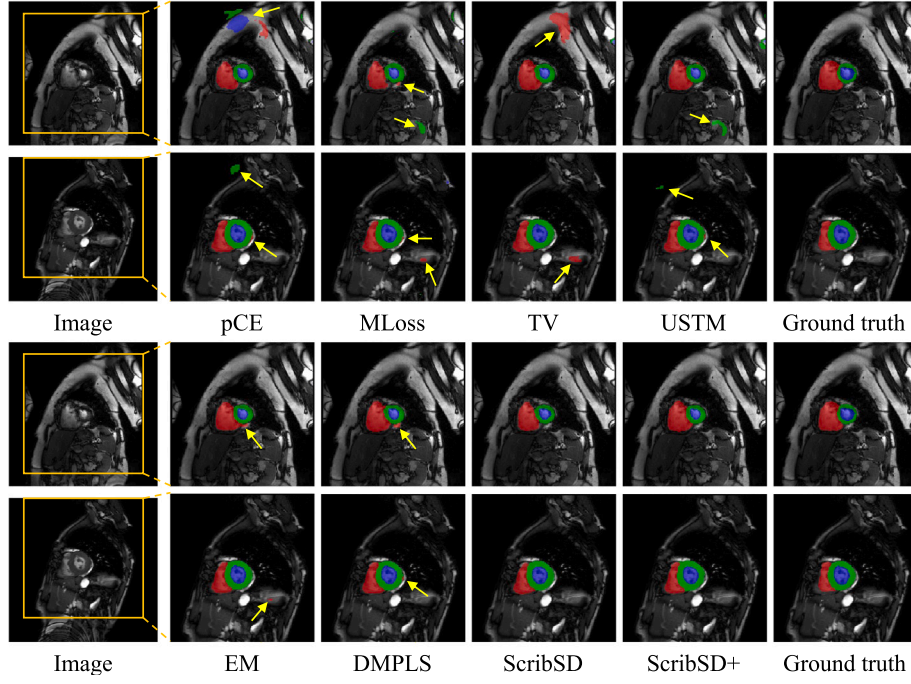


Fig. 4. Qualitative comparison of our proposed method and several existing methods on the ACDC dataset.

Table 5

Ablation study of τ with different setting strategies on the fetal MRI dataset.

Setting strategies of τ		Dice (%) $\uparrow$			ASSD (mm) $\downarrow$		
		Placenta	Brain	Average	Placenta	Brain	Average
Fixed value	$\tau = 2$	86.37 $\pm$ 3.68	89.17 $\pm$ 7.24	87.77 $\pm$ 4.80	2.92 $\pm$ 1.32	3.49 $\pm$ 4.04	3.21 $\pm$ 2.20
	$\tau = 5$	85.33 $\pm$ 4.42	90.68 $\pm$ 5.31	88.00 $\pm$ 4.24	3.12 $\pm$ 1.28	1.91 $\pm$ 1.26	2.51 $\pm$ 0.88
Cosine ramp down	$\tau_{max} = 5, \tau_{min} = 1$	86.33 $\pm$ 4.02	91.03 $\pm$ 5.07	88.68 $\pm$ 3.80	3.10 $\pm$ 1.48	2.00 $\pm$ 1.30	2.55 $\pm$ 1.10
	$\tau_{max} = 5, \tau_{min} = 2$	86.60 $\pm$ 3.59	92.16 $\pm$ 3.59	89.38 $\pm$ 3.02	3.03 $\pm$ 1.40	1.64 $\pm$ 0.86	2.33 $\pm$ 0.86

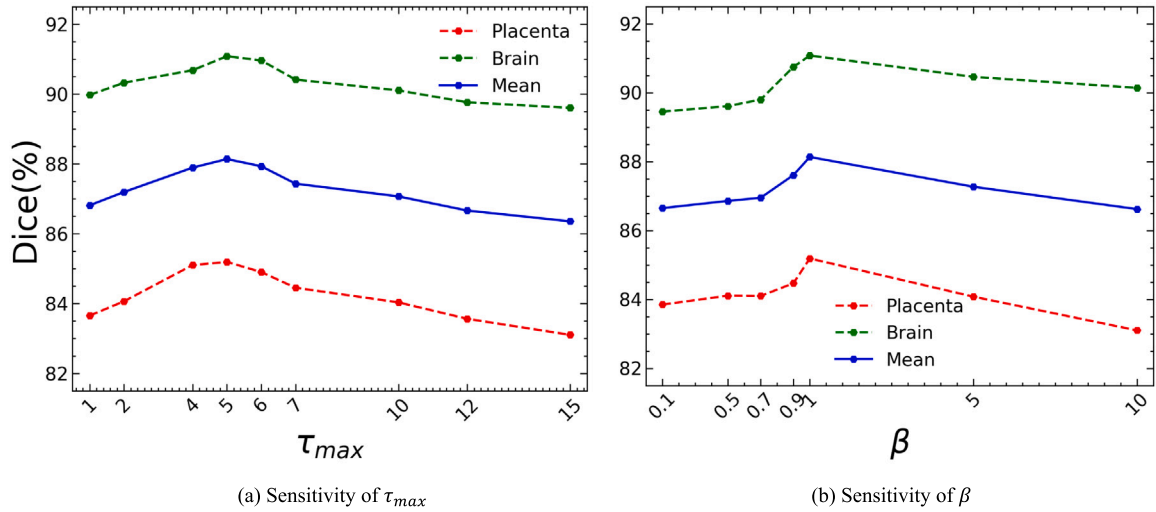


Fig. 5. Sensitivity analysis of hyper-parameters τ_{max} and β respectively on the fetal MRI dataset.

(CR) further improved average Dice to 88.27%. Finally, compared with single-scale PD and CR, combining CRF loss with our Ms-PD and Ms-CR led to the best average Dice of 89.38% with ASSD of 2.33 mm, showing the effectiveness of the proposed multi-scale prediction distillation and class-wise contrastive regularization.

4.4.2. Sensitivity analysis of hyper-parameters

In order to explore the impact of multi-scale PD and CR on model segmentation performance, we set the value of D from 1 to 5 to conduct ablation experiments. For each value of D , γ_d is the weight at scale d , and its sum is 1.0. As a smaller d corresponds to a higher resolution,

we made γ_d gradually decrease as d increases empirically. The results of different settings of the hyper-parameter D on the validation set are shown in Table 4. $D = 2, 3, 4$ performed better than $D = 1$, showing that supervision at multiple scales helps the model learn more comprehensive information. The best performance was obtained when $D = 3$. And $D = 5$, the model segmentation performed the worst. This may be due to that the bottleneck does not include sufficient spatial details for segmentation tasks.

For the ablation experiment of τ , we first investigated different methods for setting τ : (1) Setting it as a fixed value of 2 and 5 respectively, and (2) Cosine ramp down that gradually decreases τ from $\tau_{max} = 5$ to $\tau_{min} = 1$ and $\tau_{min} = 2$, respectively. The results shown in Table 5 demonstrate that our method using the cosine ramp down strategy performed better than using a fixed τ value. This is because the teacher's performance is poor in the early stage of training, and soft labels obtained through a large τ are more robust to noise. As training proceeds, the teacher's performance gets better, and the soft labels obtained with smaller τ at this time contain more accurate class member information. Table 5 also shows that $\tau_{min} = 2$ performed better than $\tau_{min} = 1$, which is mainly due to that $\tau_{min} = 1$ corresponds to a standard Softmax that is less noise-robust for knowledge distillation. Then we investigated the effect of τ_{max} by setting $\tau = \tau_{max}$ for different τ_{max} values. Fig. 5(a) shows that the performance gradually improved when τ_{max} changes from 1.0 to 5.0, demonstrating that distillation with a softer prediction based on τ -Softmax is more effective than a standard Softmax ($\tau = 1$) considering the noise in the teacher's output. However, when $\tau > 5.0$, the performance deteriorated, which is mainly because an extremely soft (i.e., highly uncertain) prediction is less informative about the class membership of pixels.

We then investigated the sensitivity of β that controls β_i in Eq. (11). Fig. 5(b) shows that the highest Dice was achieved when $\beta = 1$. Note that poor performance was obtained when setting β to very large values, which enforces the student to over-fit the teacher and limits its performance.

5. Discussion and conclusion

Our ScribSD+ framework is an end-to-end method for scribble-supervised medical image segmentation that combines the advantages of multi-scale prediction distillation and class-wise contrastive regularization. First, for multi-scale prediction distillation, we use the soft predictions obtained by the teacher to supervise the training of the student at multiple scales because soft predictions are more robust to noise and more informative. It is superior to some methods that directly use hard pseudo labels to supervise training without considering the influence of noise. For example, DMPLS (Luo et al., 2022a) generated hard pseudo labels by randomly mixing the predictions of two decoders, which will limit the learning process due to the influence of noise. Secondly, for multi-scale class-wise contrastive regularization, we extract class-wise features from feature maps and their corresponding soft predictions, constructing positive and negative pairs between the teacher and student based on different classes. We then supervise the student's training using contrastive loss, aiming to learn a more compact and distinctive class feature distribution. In addition, compared to classical KD methods (Hinton et al., 2015; Xu et al., 2020) that usually require pre-training a teacher model, we employ a self-distillation approach that effectively updates the teacher in real-time using the EMA of the model parameters.

This work still has some limitations that could be addressed in the future. First, in terms of computational efficiency, due to the use of a teacher and a student network with multi-scale predictions, our method has more computational cost and memory consumption during training than methods using a single model. For example, the training time of our method was 42 min on the fetal MRI dataset, while that for pCE was 25 min. The GPU memory consumption of our method and pCE was 9655 MB and 5629 MB on the fetal MRI

dataset, respectively. However, at the inference stage, our method only uses the student model and obtains predictions only for the highest resolution level, which makes it have the same inference time as that for pCE. The average inference time across a volume was 3.2 s and 2.1 s on the fetal MRI and ACDC datasets, respectively. Secondly, the value of τ in the τ -Softmax needs to be determined based on the effect of the ablation experiments on the validation set, which is relatively cumbersome. It would be more efficient to explore the value of τ that changes adaptively as training progresses based on available information (e.g., features or predictions) (Xu et al., 2020). In addition, we employed 2D UNet for slice-level segmentation in this work. Actually, our proposed framework is not dependent on any specific backbone network. It is of interest to use other types of backbone networks, such as 3D UNet and TransUNet, for segmentation tasks.

In summary, we proposed a self-distillation method (ScribSD+) for weakly supervised medical image segmentation based on scribble annotations. The proposed KD strategy combines multi-scale prediction distillation based on soft predictions to deal with noisy pseudo labels and multi-scale class-wise contrastive regularization that enables the model to learn more comprehensive and compact feature distribution information through class-wise contrastive loss. Our method was evaluated on the public ACDC dataset for heart structure segmentation and a fetal brain and placenta segmentation dataset. Experimental results demonstrated the superiority of our method over five state-of-the-art scribble-supervised methods. It suggested that our method has the potential for achieving accurate segmentation models based on scribbles, which can largely reduce the annotation cost for developing deep learning models in computer-aided diagnosis systems.

CRedit authorship contribution statement

Yijie Qu: Writing – original draft. **Tao Lu:** Writing – review & editing. **Shaoting Zhang:** Writing – review & editing. **Guotai Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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